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Purpose

The purpose of this document is to provide an overview of the end-to-end process of Deviation Management within the Global Deviation Database (GDD), as well as an overview of Deviation Management governance and oversight expectations. Additionally, this document covers the methods of monitoring deviations across the organization. Finally, this overview provides a consistent, global approach to the management of deviations and is aligned to the requirements set out in [LQS104, Quality Deviation Management](#)

Scope

Deviations are defined as a departure from approved processes, procedures, instructions, specifications, established standards, GxP, or regulatory requirements. This is inclusive of an unexpected occurrence where there is no procedural response.

This quality procedure applies to all GxP and submission relevant activities / quality systems within Lilly, and documentation of external deviations entered into the Global Deviation Database (GDD).

Note: Some regulations, guidance, and industry practices describe the requirements for investigating discrepancies, nonconformities, or non-conforming product. The management of these investigations is synonymous with the deviation management system as described in this procedure.

This procedure applies to activities related to product and data quality, and safety and efficacy information, for pharmaceuticals for which Lilly shares or is responsible from a legal or regulatory level, including but not limited to:

- Associated data & information
- Good Laboratory Practice (GLP)
- Good Vigilance Practice (GVP)
- Good Clinical Practice (GCP)
- Good Distribution Practice (GDP)
- Good Manufacturing Practice (GMP)
- Regulatory authorization

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1.0 Deviation Management and Lab Investigation High-Level Process Flow

The following diagram depicts a summary of the Deviation Management Process.

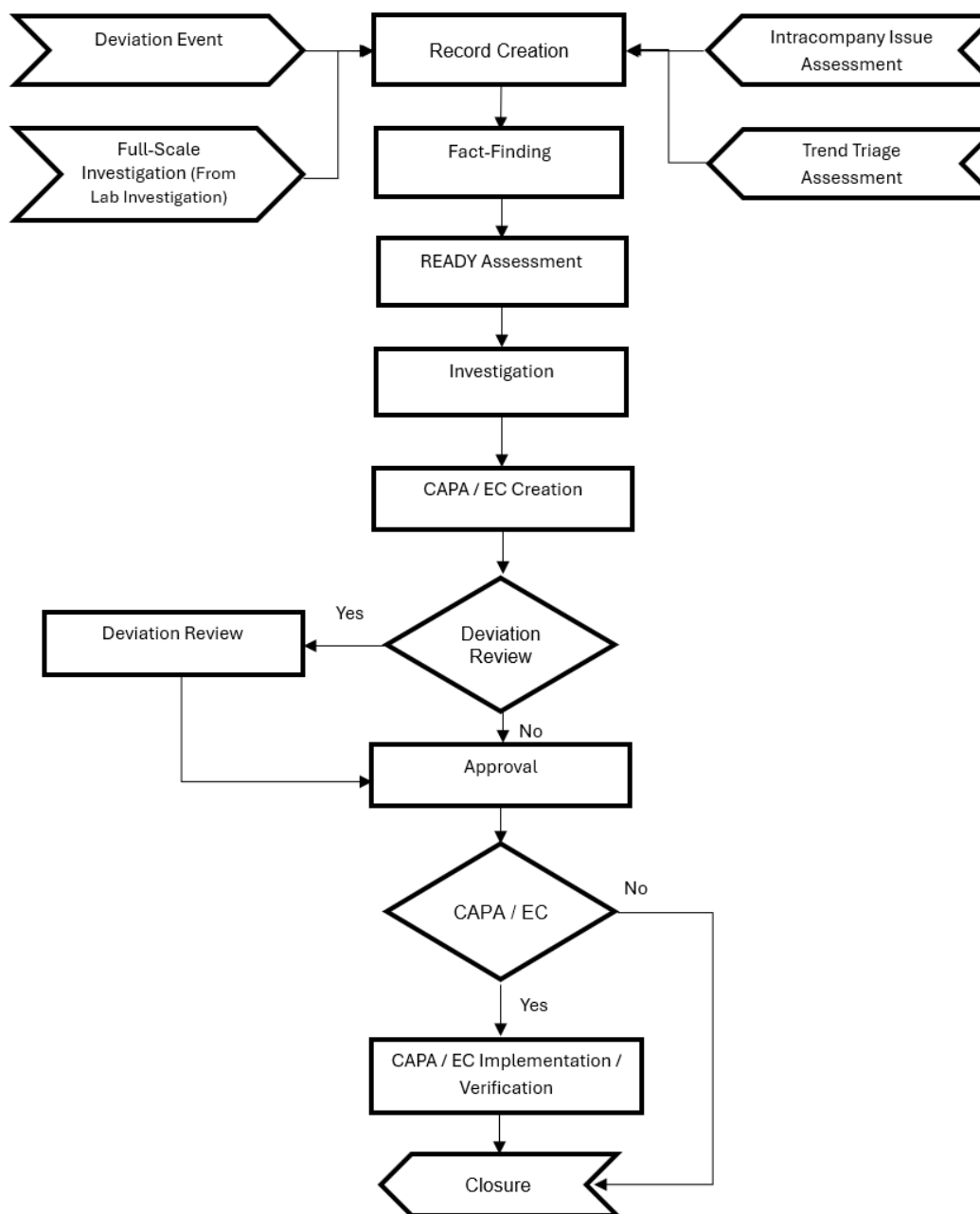


Figure 1: Deviation Management Process Overview

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1.0 Deviation Management and Lab Investigation High-Level Process Flow

The following diagram depicts a summary of the Lab Investigation Process.

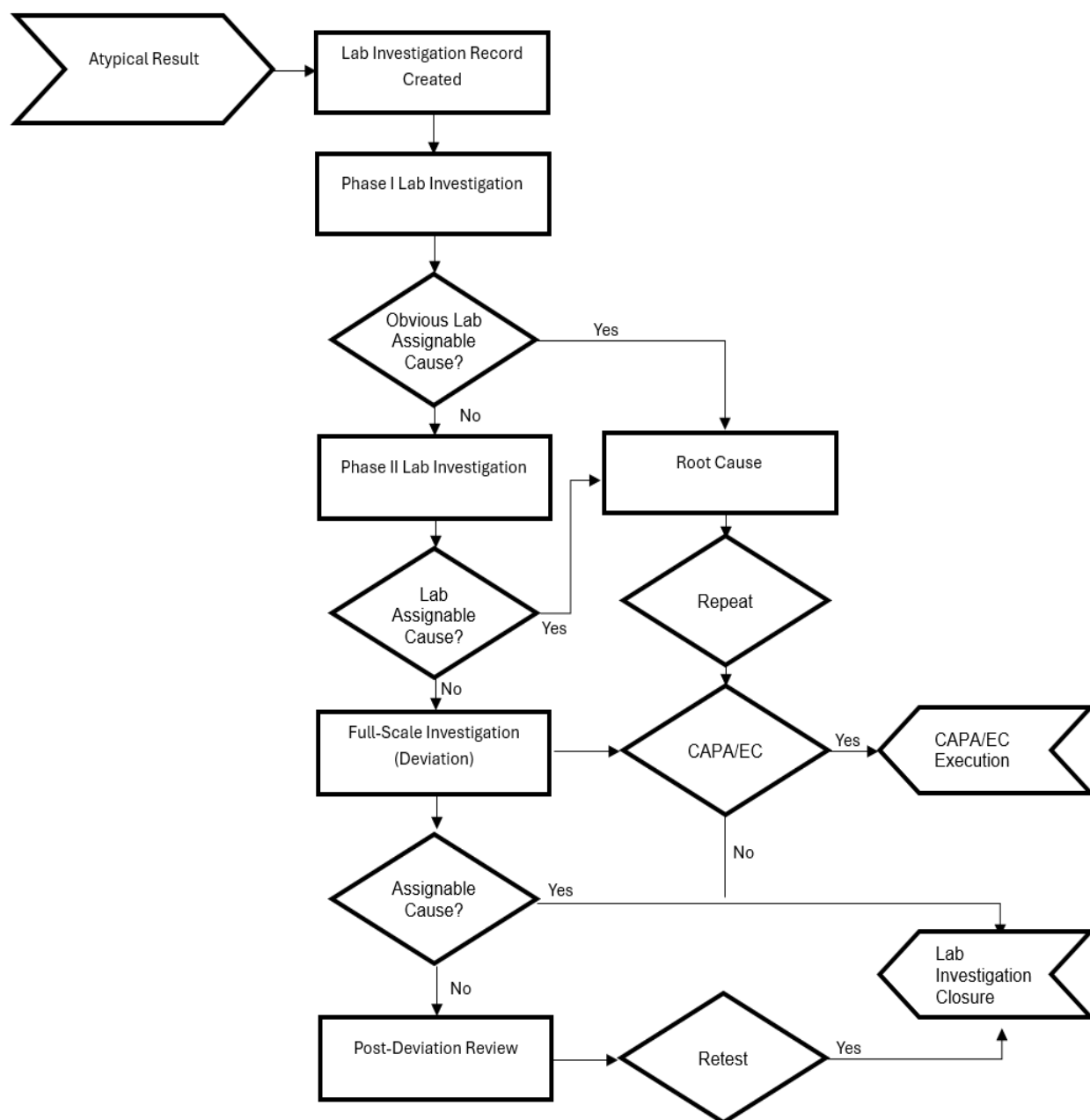


Figure 2: Lab Investigation Process Overview

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2.0 Deviation Management Governance and Oversight

2.1 Individual Deviation/Lab Investigation Oversight

Area management must ensure that deviations are identified, reported, investigated, documented, approved, and required Corrective Actions/ Preventive Actions (CAPAs) are completed. The quality unit ensures that deviations are identified and reported appropriately and CAPAs are identified/ completed, and effectiveness checks are performed to verify actions are effective in eliminating root cause(s).

Deviation review is an independent review which takes place on major and moderate deviation records where an in-system review is required. This review is performed on individual records and provides Lead Investigators with feedback on the record to ensure the deviation record is assessed and classified as required and that the investigation and appropriate CAPAs/Effectiveness Checks (ECs) have been created.

Deviation approvers (quality assurance (QA) and general) provide feedback and oversight to individual records and are responsible for ensuring that deviation records are accurate, complete and comply with all requirements per [LQS104, Quality Deviation Management](#), prior to approving a record.

Table 1 lists the roles and responsibilities for users executing deviation records. Users of the deviation management system must be suitably trained for the tasks they are performing.

Table 1: Roles and Responsibilities

Deviation Role	Responsibilities
Initiator	Create deviation or laboratory investigation records within the GDD and complete preliminary information within the record.
Fact-Finder	Document all initially available facts about the deviation which are available at the time the record was initiated. This includes, but is not limited to, times, locations, impacted areas, objects, batches and products.
READY Assessor	Perform assessments on deviation records, which includes the quality risk assessment and classification of deviation records by use of the Risk Priority Number (RPN) tool. Must be a quality representative for all deviations.
Lead Investigator	Complete all required investigation activities, including the root cause analysis. Create and assign child records, such as child investigations and user tasks. Ensure all child records are complete. Complete the summary of the investigation. Create associated CAPAs/ECs.
Reviewer	Review deviation records for completeness and accuracy after the completion of an investigation. Provide feedback and determine if revisions to the reviewed records are required.
Approver	Approve records by providing an electronic signature signifying the records have been reviewed for accuracy, completeness, respective to the approval role, that there is agreement with conclusions, and that appropriate CAPAs have been created, as necessary.
User Task Owner	Complete the assigned task, in accordance with the User Task instructions, by the assigned due date of the User Task.

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2.0 Deviation Management Governance and Oversight

Deviation Role	Responsibilities
Child Investigation Owner	Complete the assigned Child Investigation in accordance with the instruction and required information requested by the assigned due date of the Child Investigation.
CAPA Owner	Implement and complete the assigned CAPA as described in the CAPA record. Document outcomes and results of the implemented CAPA.
EC Owner	Execute assigned Effectiveness Checks (ECs) and provide results in accordance with the instructions and acceptance criteria provided within the Effectiveness Check record. Assign a verdict of the EC based on the results.
EC Verifier	Review and confirm the execution and associated outcomes of an EC.
One-Time Approver	Approve Private User Tasks or Private Child Investigations without reviewing or approving related deviation or laboratory investigation records.
Quality Alert Initiator	Create and document Quality Alert and Quality Alert Addendum information so that Quality Alerts can be sent to recipients to take appropriate actions.
Quality Alert Reviewer	Review Quality Alerts and Quality Alert Addendums prior to the alerts being sent. Assign a verdict to the record for approval to be sent.
Private User Task Owner	Complete the assigned Private task in accordance with the Private User Task instructions by the assigned due date.
Private Investigator	Complete the assigned Private Investigation in accordance with the instructions, and required information requested, by the assigned due date of the Private Investigation.
External Party	Complete assigned tasks as a user external to Lilly by use of an external link allowing temporary access to the assigned task for completion.
CAPA Verifier	Review and confirm the actions and associated outcomes of an implemented CAPA.
Lab Investigator	Complete all required investigation activities (Phase I, Phase II, and Full-Scale), including the root cause analysis. Create and assign child records, such as test plans, child investigations, and user tasks. Ensure all child records are complete. Complete the summary of the investigation.
Lab Investigation Quality	Complete tasks within a Laboratory Investigation record as a Quality Assurance representative.
Test Plan Owner	Complete Test Plans in accordance with the Task instructions by the assigned due date.
Impact Assessment Owner	Complete the assigned Impact Assessment in accordance with the instructions, and required information requested, by the assigned due date of the Impact Assessment.

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2.0 Deviation Management Governance and Oversight

2.2 Deviation Local Governance

Local sites and areas must apply all requirements from this procedure and reference [LQS104, Quality Deviation Management](#). It is the sites' responsibility to ensure that a local governance is in place to handle feedback from the global governance, monitor their own deviations, identify systemic issues, and implement appropriate actions to ensure continuing suitability, adequacy, and effectiveness of the deviation management system and to identify opportunities for improvement.

2.3 Deviation Global Governance

The Deviation Global Process Owner (GPO) is responsible for the deviation management process and associated system performance. The GPO ensures that changes to the Global Deviation Database (GDD) are managed appropriately and manages supporting tasks such as global document reviews/updates. The GPO monitors the overall performance of the deviation management system and escalates to appropriate levels of management when issues or opportunities are identified, and actions need to take place.

3.0 Deviation Management Process Overview

The global approach to deviation management includes the following process steps for the execution of deviation records.

3.1 Deviation Identification

All employees must be alert to indicators of potential deviations, as defined. If a deviation is suspected, an immediate response must be taken to prevent further impact, if possible. Additionally, a deviation record must be created to document the departure and will include any conditions that may adversely affect the product or associated data and be reported to immediate supervision. If a user does not have access to create a deviation in the GDD, the appropriate area representative must be contacted to create the deviation record.

The Quality Incident module contains two workflows which also serve as sources of deviation identification. The Intracompany Issue (ICI) workflow is intended to manage issues which are discovered between sites that require an assessment as to whether a deviation is required. Where the assessor determines that an investigation is required, a deviation must be created from the ICI record. If a deviation is already open which addresses the ICI record, the assessor may choose to relate this record rather than create a new one.

A Trend Triage is the second workflow within the Quality Incident module where a suspected trend or emerging theme has been identified and requires an assessment by quality to determine if an investigation is required. If the assessor determines that an investigation into the suspected trend or emerging theme is required, a deviation must be created from the Trend Triage record. If a deviation is already open which addresses the Trend Triage record, the assessor may choose to relate this record rather than create a new one.

3.2 Deviation Notification

It is required that the READY Assessor, a member of QA, be notified of deviations. This may be achieved by assigning the record to the READY Assessor in the GDD.

Refer to [GSOP-101-3, Quality Notification to Management](#), for events that qualify for further escalation.

Immediate reporting of quality issues to a regulatory authority must follow [LQS112, Regulatory Authority Communication and Inspections](#).

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3.0 Deviation Management Process Overview

3.3 Record Creation and Fact Finding

Users are required to create deviations in the GDD within one (1) business day of deviation discovery. Deviation records are assigned a unique number and must include a defined problem statement and information about the event.

Fact Finding identifies the scope of the deviation, which includes, but is not limited to batch impact, organization impact, clinical study impact, assets, and materials. Impact on patient/human research, safety, product quality, safety/efficacy data, and regulatory information must be evaluated. Where batches or products have been included or excluded from the scope of the deviation record, a justification must be documented within the deviation record. All known available data/facts about the issue should be included in the fact-finding step prior to assessment with as much information as possible to aid in the understanding of the risk associated with the deviation event and to provide the assessor with as much information as possible to appropriately assess and classify deviation records. The Fact Finder may utilize the Artificial Intelligence Virtual Mentor to provide feedback on, and recommended updates to, the deviation description and immediate actions.

For deviation records associated with a product batch, the deviation number must be recorded in the batch production record or equivalent electronic document system.

3.4 READY Assessment

The READY (Risk Evaluation, Actionable Direction, and Yield Results) Assessment is performed for deviation records (Laboratory Investigations are excluded from the READY Assessment process). A READY Assessor must consider all elements of risk associated with a deviation event. In addition to assessing risk, the assessor provides direction and guidance on how the investigation could be conducted for Lead Investigator to utilize for investigation direction (e.g., root cause analysis tools, suggested corrections, etc.).

The READY Assessor must ensure that all processing, materials, and any other objects identified as within the scope of the deviation are suitable for continued use during the investigation. Additionally, involved/impacted batches are reviewed and assessed during the READY Assessment process. All deviations are also assessed to determine if escalation of the issue is required or if there is potential impact to product on the market.

A deviation classification must be assigned to the record. Deviations are classified utilizing the Risk Priority Number (RPN) which requires users to answer questions related to severity, detectability, and occurrence of the deviation event. By answering these questions, a score will automatically be generated, and the score will be used to classify the deviation record as Major, Moderate, or Minor. Major deviations are considered the highest risk / most severe deviations. The READY Assessor has the option to provide a justification to override the RPN classification.

Deviation records may be cancelled up to the READY Assessment stage; however, cancellation may only be done by Quality and justification must be provided.

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3.0 Deviation Management Process Overview

3.5 Investigation

The purpose of the investigation is to determine and address the root cause of the deviation, identify any additional impacts not yet detected, evaluate final quality and/or regulatory impact, and define appropriate CAPA/EC. The level of effort, formality, and documentation of the Root Cause Investigation (RCI) and CAPA must be commensurate with the criticality of the deviation.

- An appropriate level of root cause investigation must be applied utilizing a risk-based approach. The approach utilized must be documented.
- The investigation must ensure that quality system elements, including process, procedural, or system-based errors or problems, are investigated first. If personnel performance is suspected or identified as a root cause justification must be provided. In cases where the true root cause(s) of the issue cannot be determined, consideration must be given to identifying the most likely root cause(s).
- Appropriate CAPA(s) & ECs must be identified and implemented.

The impact of the quality deviation must be investigated, documented, and consider the following:

- Patient or human research subject safety
- Product/data quality
- Quality and integrity of safety/efficacy information
- Regulatory information, including compliance with the approved marketing authorization

Condition: Area is a GMP/Manufacturing Site, also consider:

- Impact on stability (including the need for additional stability studies)
- Results of product and process monitoring
- Qualification or validation state of the process and systems
- Whether the deviation is systemic (e.g., effectiveness of the control strategy, adequacy of the quality system)

Upon completion of the deviation investigation, it is required that major and moderate deviations have a recurrence check performed to identify any previous deviations in the GDD reported in the same area which have the same description and root cause. Minor deviations are not required to have the recurrence check performed due to the occurrence check that is performed at READY Assessment as part of the RPN classification process. If repeat occurrence has been detected, the scoring will indicate a higher risk and may result in a higher classification level.

The investigation must extend to all impacted areas. Several tools are available to the investigator to help/ support the investigation process (e.g., root cause analysis tools, 5 whys investigations, user tasks, AI tools). Once the root cause is identified, the lead investigator is required to establish the required corrective and preventative actions to address the identified root cause and prevent recurrence of the deviation.

The lead investigator will have AI tools to aid with the investigation phase.

- The Root Cause Categorization tool will provide the user with root cause recommendations based on the investigation report.
- The Virtual Mentor for Investigation provides in-system feedback on investigation reports and offers suggestions for improvement.
- The Executive Summary Generator provides a draft executive summary for investigation reports.

These tools require the lead investigator to review the output(s) of the tool as the “human in the loop” and accept the information, or edit the information, as needed.

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3.0 Deviation Management Process Overview

3.6 Review and Approval

Deviation review must be performed as an independent review by one or multiple people after the completion of an investigation and prior to deviation record approval for deviations where the system requires mandatory review. The deviation review is intended to ensure the deviation was appropriately investigated, the correct root cause(s) was identified, and appropriate CAPAs/ ECs were identified and created. The review step is captured within the GDD by the reviewer providing feedback and indicating to the lead investigator whether updates are required to the record or not.

Deviation record approval is done as the final step of the deviation execution process. There are a minimum number of approvers (business and quality) and minimum levels of approvers for each deviation record, depending on the deviation classification and area related to the deviation. Once approval is completed, CAPA(s) implementation begins, or the deviation record will close. QA must ensure approval of the deviation record prior to release of impacted batches, materials, etc. Internal deviation records are required to be approved/ completed within 30 calendar days of the date discovered. External deviation records are required to be approved/ completed within 45 calendar days of the date discovered. Where a deviation record does not meet the 30-day or 45-day approval period, an extension request must be created and approved prior to the due date of the deviation. Where a record has been approved or completed, and an extension request was required, an amendment must be created to address the missed deadline and associated risk. At minimum, Quality must approve all extension requests.

If minor additional information (for example, changes to the conclusion or rationale) becomes available after approval of the deviation record, the impact of this information must be evaluated, and updates must be made to the record to include additional information or reference to this information via the Amendment process. Any modification of a deviation record's content following approval, including the batch disposition decision, planned corrective action, preventive actions, or effectiveness checks must be re-approved by the original approvers or their equivalents. Approved or completed records cannot be reopened, and the Amendment process must be followed.

3.7 CAPA and EC

Corrections are identified activities that are put in place to mitigate a deviation event, and which do not address root cause. Corrections are created within the deviation record and are to be implemented prior or after deviation record approval.

Corrective and Preventive Actions (CAPAs) are identified and created to address identified root cause(s). It is required that CAPA(s) is created or existing CAPA(s) are added to major and moderate deviation records, and the deviation root cause record is linked to the associated CAPA. CAPA(s) are implemented after deviation record approval. If it is determined that no CAPA is necessary for a major or moderate deviation, justification must be provided.

For CAPAs related to medical devices and device constituents of combination products, there must be a verification performed to confirm that the action does not adversely affect the ability to meet applicable regulatory requirements or the safety and performance of the medical device. This must be evaluated prior to CAPA plan approval and during the verification of effectiveness.

CAPA and Corrections must not be investigational. Where an assessment of scope of impact is required to assign actions, that assessment must be completed prior to investigation approval.

It is required that all created CAPA(s) have a related EC created to ensure the implemented CAPA(s) has addressed the identified root cause(s) and that the implemented action will further prevent recurrence of the deviation. All CAPA require verification of implementation by Quality. EC verification by Quality is required for major and moderate deviations. Appropriate actions must be taken to address any EC which has been determined as ineffective or inconclusive.

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3.0 Deviation Management Process Overview

3.8 External Deviations

External deviations are those that are created and managed by external parties (including External Laboratories) and are entered into the internal GDD for tracking and investigation, if required in accordance with established agreements with external parties. External deviations are required to be approved/ completed within 45 days of discovery (notification). The External Collaboration feature may be used with external parties allowing them to provide information directly into the deviation record utilizing a secure link.

3.9 Privacy

Consideration must be given to deviation of data that requires privacy. Deviation data that requires privacy must be documented in a Private User Task or Private Investigation, and includes the following examples:

- Personnel information such as an individual's name/ID when documented in support of a personnel performance issue, and/or,
- In limited cases and in alignment with management decision, when other root causes human error have been identified and personnel performance issue is a significant contributing factor, and/or
- Information of a sensitive nature (e.g., personal problem/physical issue), and/or
- Investigation details that identify a personnel performance issue, human error with personnel performance issue or as instructed by management.

3.10 Actions in Response to Nonconforming Product Detected before Delivery

Nonconforming product must be dispositioned by taking one or more of the following actions:

- eliminate the detected nonconformity,
- preclude its original intended use or application, or
- authorizing its use, release or acceptance under concession.

Ensure that nonconforming product is accepted by concession only if the justification is provided, approval is obtained, and applicable regulatory requirements are met. Records of the acceptance by concession and the identity of the person authorizing the concession must be maintained.

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4.0 Laboratory Investigation Process Overview

4.1 Laboratory Investigations

Laboratory Investigations are investigations conducted to address an Out of Specification (OOS), Out of Trend (OOT), or Out of Expected Result (OOE) and are managed through a separate workflow within the Global Deviation Database. The workflow is configured to the specific steps and requirements for conducting Laboratory Investigations including Phase I, Phase II, and Full-Scale (deviation) investigations.

Any result which is considered an atypical result (OOS, OOT or OOE) must be investigated. Any confirmed OOS result, or significant negative trend, affecting product batches released on the market should be reported to the relevant competent authorities. The possible impact on batches on the market should be considered.

Laboratory Investigations are required to be approved within 30 calendar days of discovery. If a deviation record is related to a Laboratory Investigation, use the same discovery date of the Laboratory Investigation. The related deviation record will have 30 calendar days to complete.

If minor additional information becomes available after approval of the Laboratory Investigation record, the impact of this information must be evaluated, and updates must be made to the record to include the additional information or reference to this information via the Amendment process. Any modification of a Laboratory Investigation record's content following approval, including the sample validity decision, planned corrective action, preventive actions, or effectiveness checks must be re-approved by the original approvers or their equivalents. Approved or completed records cannot be reopened, and the Amendment process must be followed.

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4.0 Laboratory Investigation Process Overview

4.2 Laboratory Investigation Identification

Whenever possible, analysts must check the sample data for compliance with test specifications before discarding test preparations or standard preparations. Sample preparations must not be discarded before second person verification has been completed for investigative purposes. Once an OOS, OOT, or OOE has been identified, (i.e., following second person verification), a laboratory investigation must be initiated, and area supervision and quality must be notified.

For chemical methods, the following must be investigated as an OOS result:

- If one or more of the replicates are outside specification, where the test methodology comprises several replicates, and the result is within specification, or
- Any OOS results generated, even when method system suitability criteria or method acceptance criteria were not met.

Due to the unique nature of animal, biological, and microbiological methods, the definition for what constitutes OOS or OOE must be defined by the specific analytical test method.

The OOS/OOT/OOE result obtained must be logged in an electronic or hardcopy tracking system and must include the following information:

- Test
- Product/material
- Batch number
- Laboratory Investigation number
- Date

Upon identification of an OOS or OOT stability result, the person with responsibility for stability study oversight must be notified immediately. This individual must determine the status of the study and make a recommendation to the site quality leader to either continue or discontinue the laboratory investigation and/or deviation investigation according to the laboratory investigation requirements.

4.3 Phase I Lab Investigation

Once an out of specification, out of trend, or atypical result is identified, a Phase I Lab Investigation is performed for the purpose of identifying any obvious or immediate assignable cause. If no obvious assignable cause is identified, an appropriate Phase II Lab Investigation must follow.

4.4 Phase II Lab Investigation

If an assignable cause cannot be immediately identified during the Phase I Lab Investigation, a Phase II Lab Investigation must be conducted. The supervision of the lab area must ensure this is completed and that in addition to a Quality approver, a technical approver is also involved. The technical approver must ensure that all investigation activities are documented, and scientific rationale is applied throughout the investigation process. Investigational testing may be used to confirm or disprove a hypothesis or possible assignable cause. If the analytical investigation determines an assignable cause, the original results are invalidated and a repeat test, as appropriate, is performed.

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4.0 Laboratory Investigation Process Overview

4.5 Full-Scale Investigation (Deviation)

If no assignable cause is identified during the Laboratory Investigation, the investigation may be extended outside of the testing area into manufacturing by means of a deviation. A deviation may also be initiated at any time prior to the completion of the laboratory investigation, if warranted by circumstances.

If an assignable cause is identified during the deviation, the deviation and laboratory investigation record(s) are completed and approved. If the deviation does not identify an assignable cause, a retest may be performed to confirm the OOS or OOT result.

If either the original sample was consumed during the investigation process or the laboratory investigation or deviation investigation determines that the original sample integrity was compromised, then a resample may be obtained to perform a repeat test or a retest.

Quality and technical approvers are required to approve the full-scale investigation at the point of the deviation review process within the Laboratory Investigation record.

4.6 Test Plans

Test plans are utilized for performing tests to support laboratory investigations and/or deviation investigations. Certain types of tests may be performed under certain circumstances.

A retest of the original sample may be conducted when no assignable cause can explain the result through a deviation investigation. Retesting may only be permitted to be performed on a sample one time. Any departure from this requirement must be approved by the site quality leader.

A resample is only permitted when the original sample is consumed, or there is a proven issue with sample integrity. If a resample is required, then the new sample must be taken in the same manner as the original sample unless the original sampling method was determined to be incorrect. The difference from the original sample method must be documented and justified.

5.0 Deviation/Laboratory Investigation Monitoring

Deviation and Laboratory Investigation monitoring provides a structured approach for monitoring and analyzing deviation and laboratory investigation data.

The program consists of two main elements:

1. **Emerging Themes** – This monitoring method utilizes data review to identify unexpected patterns, , or anomalies in deviation data. This process helps to uncover hidden or unknown risks derived from occurrence checks, recurrence checks, and ad-hoc monitoring.
2. **Performance Monitoring** – This monitoring method utilizes data analysis of predetermined parameters to ensure continuing suitability, adequacy, and effectiveness of the deviation management system and to identify opportunities for improvements. Performance monitoring must be performed and is reported and reviewed as defined by site/ area/ function and global trending.

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5.0 Deviation/Laboratory Investigation Monitoring

5.1 Emerging Themes

Emerging theme monitoring utilizes data review to identify unexpected patterns, correlations, or anomalies in deviation data.

Emerging themes are monitored during the execution of each deviation record through the following checks:

- **Occurrence Checks:** Performed at READY Assessment on all deviation records to monitor and identify risk associated with deviations where there have been repeat instances of a deviation event. By utilizing the occurrence checks, emerging themes can be detected early, and the risk of occurrence will be considered with the RPN tool, possibly elevating the deviation classification.
- **Recurrence Checks:** Performed on moderate and major deviation records which provide a report on the number of times a similar deviation has happened in the same area with the same root cause. When recurrence has been identified, a user must state whether action is required because of the recurrence. If the scope of the existing deviation or existing CAPA adequately addressed the recurrence, no further investigation is required. A user may open a Quality Incident to perform an assessment to determine if a new deviation is required to address the recurrence.

Additional types of monitoring, such as retrospective reporting (for example, Periodic Deviation Monitoring Report) and ad hoc queries may also be used to identify emerging themes across deviations and laboratory investigations. This type of monitoring may be performed on a periodic basis, or as quality or other functions deem appropriate. If any other type of monitoring identifies unexpected patterns, correlations, or anomalies in deviation data, a Quality Incident (QI) record must be used to perform an assessment to determine if a new deviation is required to address the emerging theme.

5.2 Performance Monitoring

Performance monitoring is assessed using the Periodic Deviation Monitoring Report (PDMR). Local governance is responsible for establishing performance monitoring. Metrics and data are required to be analyzed periodically by sites/functions to monitor performance of deviation management and to document suspected trends within the Quality Incident module of the Global Deviation Database (GDD) to assess if further investigation (i.e., a deviation) is required.

Additional performance monitoring on a global level serves to identify trends and process improvements. This monitoring also ensures appropriate action is taken and drives continuous improvement within the deviation management system.

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Revision History

The Revision History will only include the current version of the document changes.

Refer to previous document versions and/or revision history metadata in the electronic document management system (eDMS) for historical changes.

Version Number	Effective Date	List of Major Changes
2	Effective date viewable in eDMS.	Section 3.1 changed wording from “users” to “employees”. Clarified instructions in section 3.2 to refer to GSOP-101-3 for events that qualify as a Notification to Management. Section 3.9 corrected second bullet from causal factor to root causes. Section 4.1 clarified second paragraph that any result that is considered atypical must be investigated. Clarified that Quality Deviations that are linked to laboratory Investigations date of discovery is not automatically added, but they need to be given the same date of discovery and removed who is completing the investigation. Clarified that the Laboratory Investigation module will only address sample validity decision. In section 4.4, reworded from “discount” to “disprove”. Modified section 5.0 to align with revision of GSOP-104-7, Deviation Monitoring Program. Included GSOP-104-7 in the Related Procedures section. Removed Amendment instructions from section 4.2, as it is already included in section 4.1.

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Appendix A – Related Procedures

Procedure	Intent
GSOP-104-1 Record Creation & Fact Finding	Process and requirements for users to create deviation records within the Global Deviation Database (GDD) and perform Fact Finding, including adding impacted items, such as batches, to the deviation record.
GSOP-104-2 READY Assessment	Process and requirements of identifying risk and performing the classification of deviation records by using the Risk Priority Number (RPN) tool.
GSOP-104-3 Investigation	Description of the process of conducting deviation investigations, including the identification of Root Cause(s), CAPAs/ECs, and use of AI tools to help aid in investigations.
GSOP-104-4 Review and Approval of Deviations	Requirements and process of performing Deviation Reviews and approvals of deviation records.
GSOP-104-5 External Deviations	Requirements and execution of deviation records which involve external parties, including the functionality of the External Collaboration feature, allowing external parties to contribute to deviation records, as needed.
GSOP-104-6 Laboratory Investigation	Process of identifying and conducting Laboratory Investigations, including Phase I, Phase II, and Full-Scale Investigations (related deviations).
GSOP-104-7 Deviation Monitoring Program	Process to monitor events in the GDD, to assess the health of the Deviation Management Program.
GSOP-101-1 CAPA and Effectiveness Checks	Process and requirements of creating CAPAs and Effectiveness Checks (ECs) and the process of CAPA implementation, EC execution, and EC verification.

Document Approval Signatures

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